

System Design

A decorative dashed line forms a large rectangle. At the top-right corner, a dashed arrow points upwards and to the left. At the bottom-left corner, a dashed arrow points downwards and to the right.

Bulkhead Pattern

\$ whoami

Matheus Fidelis

Engenheiro de \$RANDOM

@fidelissauro

<https://fidelissauro.dev>

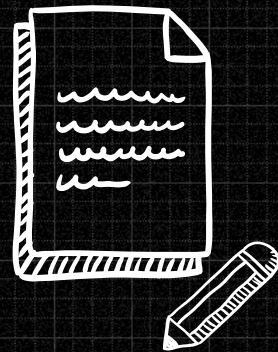
<https://linktr.ee/fidelissauro>

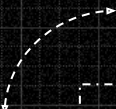


OBJETIVOS

- ✓ Visão Geral
- ✓ Recap de Aulas Anteriores
- ✓ Tipos de Bulkheads
- ✓ Estratégias
- ✓ Arquiteturas de Referencia

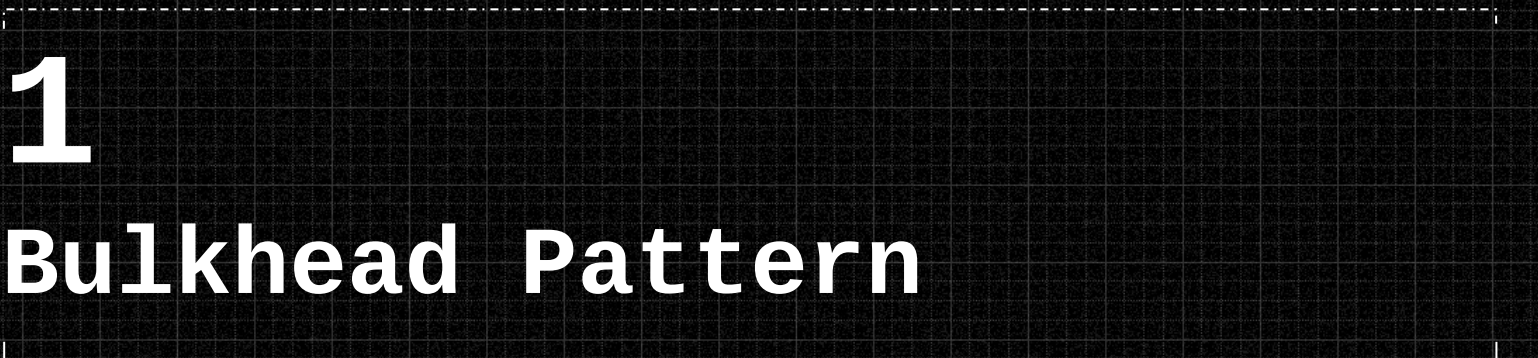
RECADOS



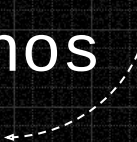


1

Bulkhead Pattern



Objetivos e
Ganhos

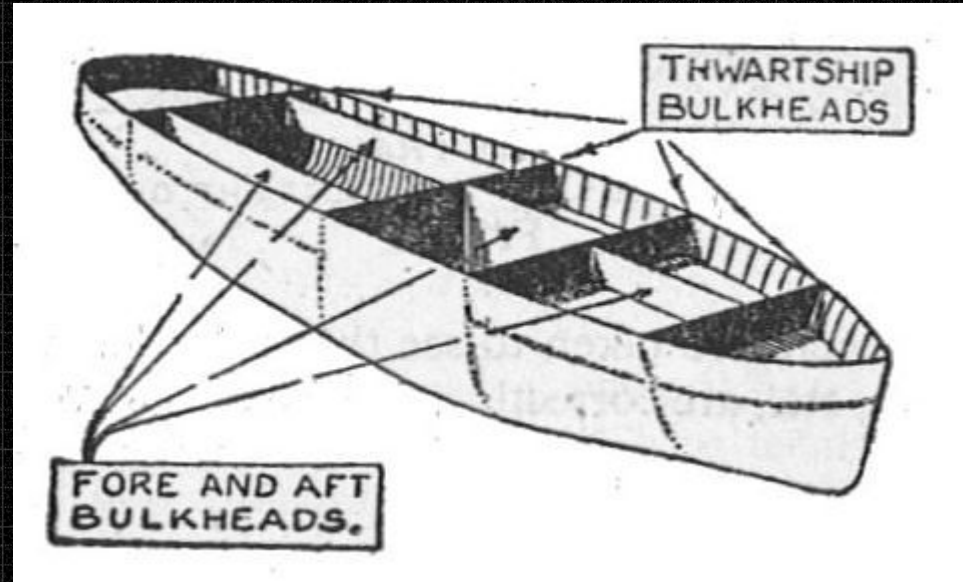


Bulkhead Pattern

- Bulkhead como estratégia de contenção de falhas (não de prevenção)
- “sistema homogêneo” para “compartimentos”
- Destinos operacionais distintos
- Níveis de Bulkhead:
 - lógico → físico → sharding)

Origem na Engenharia Naval

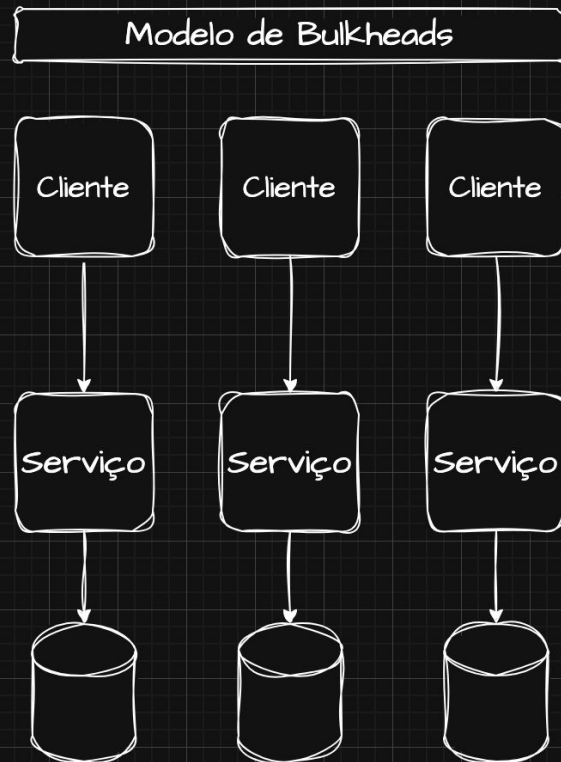
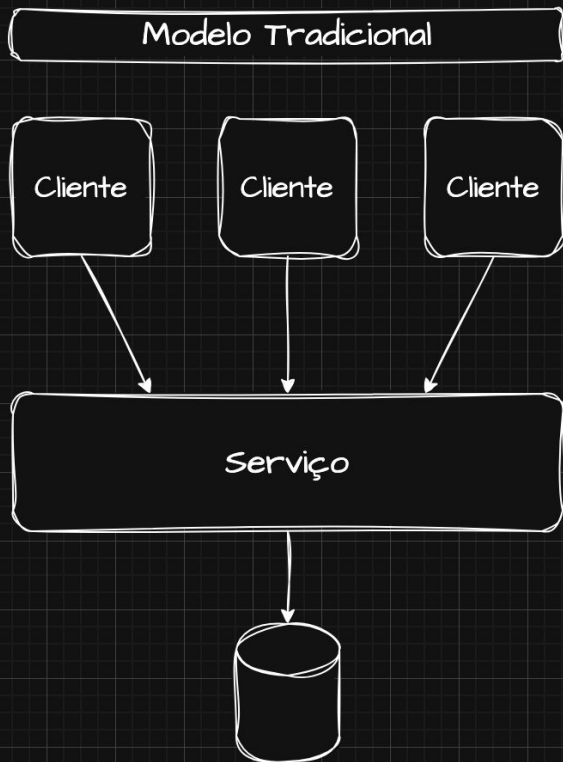
- Engenharia Naval
- Bulkheads no casco
- Contenção
- Impedir Naufrágios



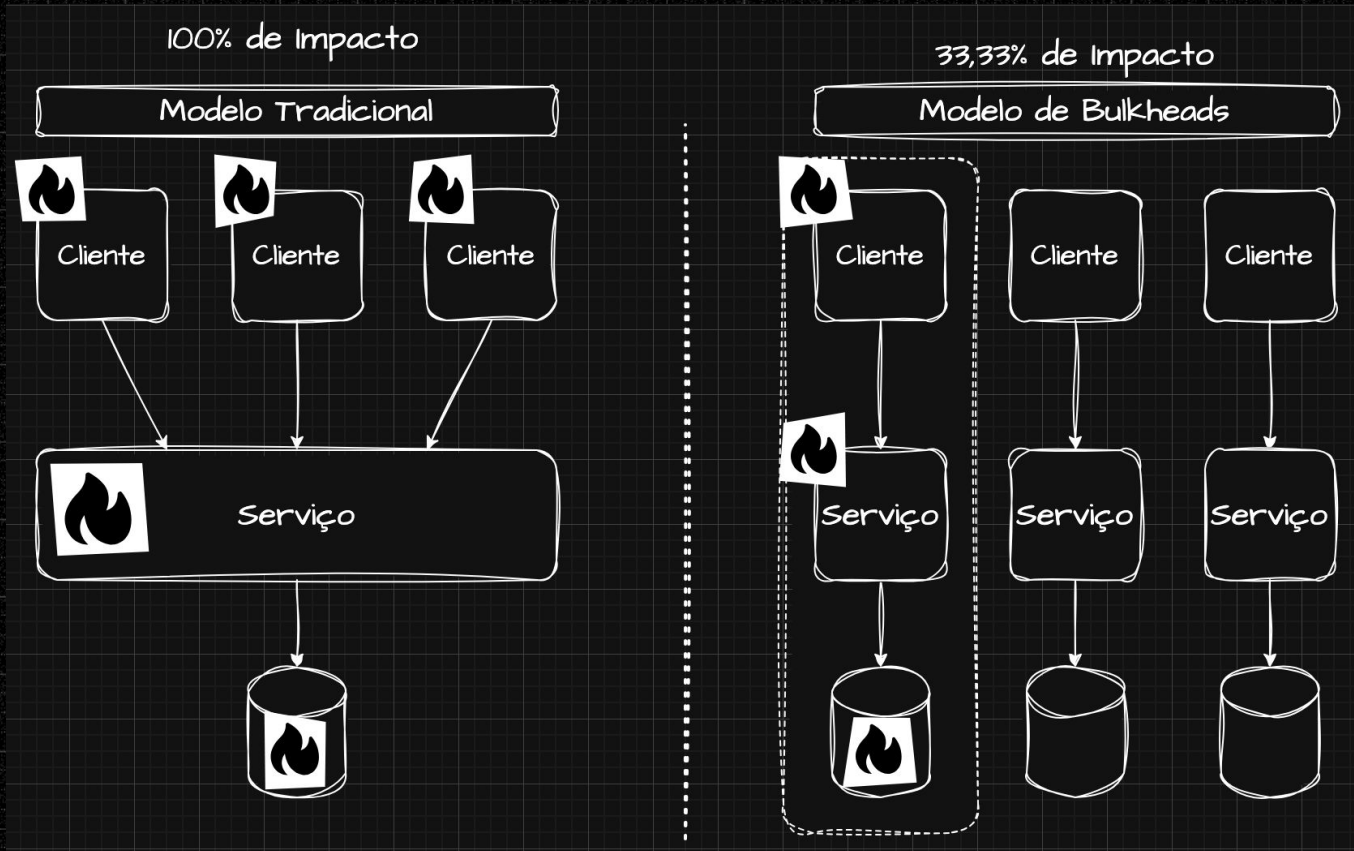
Bulkhead na Arquitetura de Software

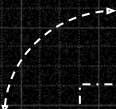
- Separação de recursos e destinos de execução
- Se compartilha o mesmo recurso finito, compartilha o mesmo destino
- Pode existir em múltiplas dimensões:
 - Threads,
 - Filas,
 - Pools de conexão,
 - Databases,
 - Clusters,
 - Shards

Bulkhead na Arquitetura de Software

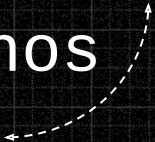


Bulkhead na Arquitetura de Software





2 Bulkheads Lógicos e Físicos



Objetivos e
Ganhos

Implementação

- Bulkheads podem ser implementados em diferentes níveis da arquitetura
- A implementação correta exige clareza sobre quais recursos são finitos e como eles devem ser particionados
- Recursos Lógicos
- Recursos Físicos
- Sharding Computacional

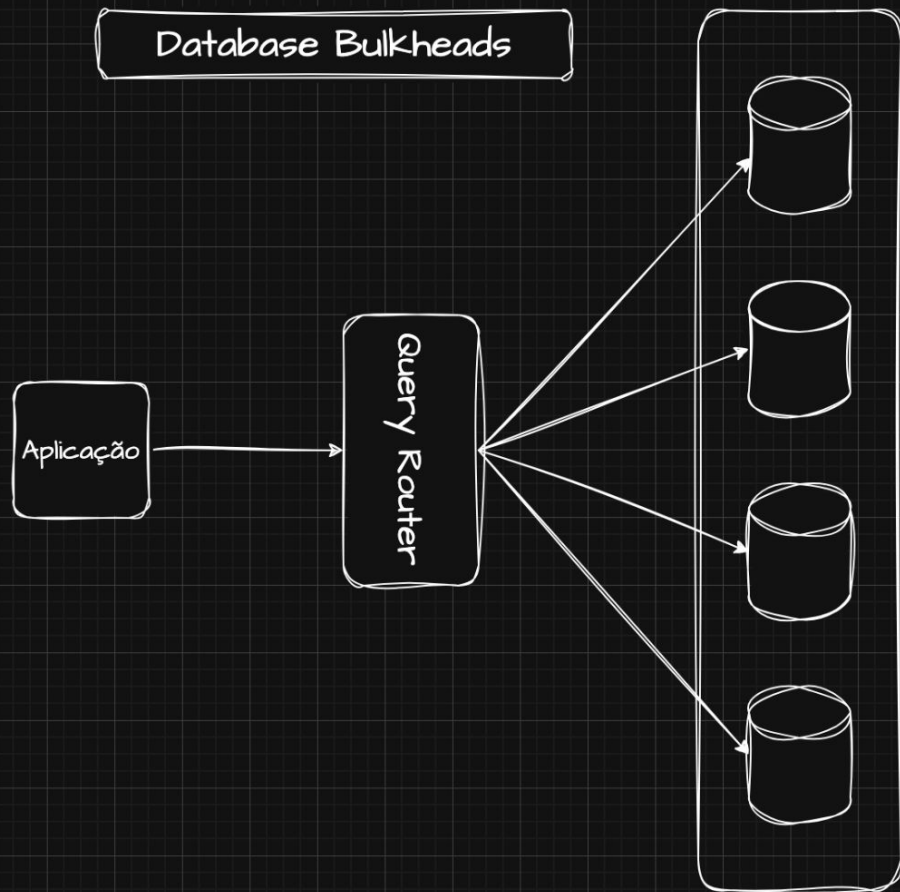
Recursos Lógicos

- Recursos de execução e concorrência
- Thread Pools
- Connection Pools
- Semaforos
- Rate Limiting

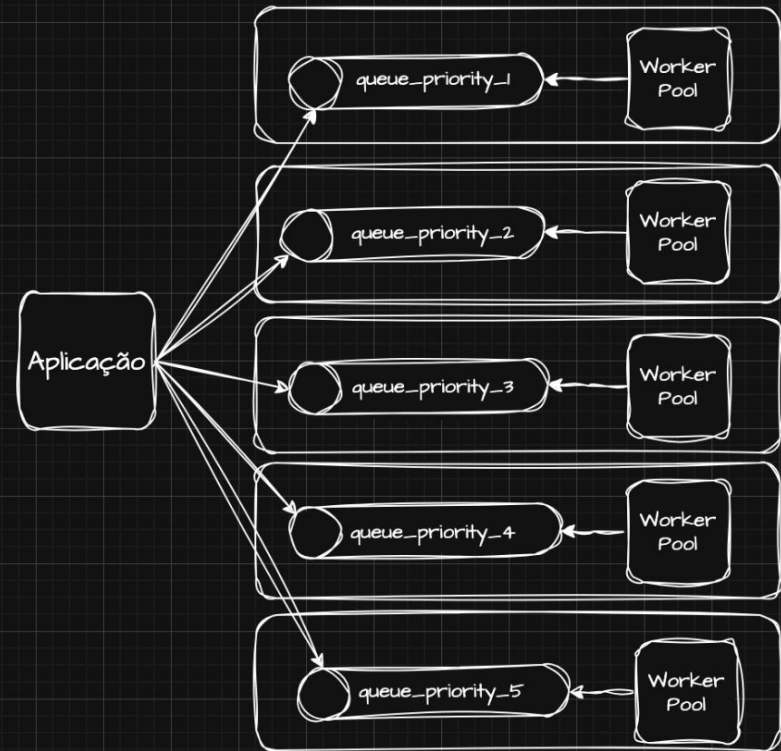
Recursos Físicos

- Separação concreta de infraestrutura
- Segregar e Dedicar recursos físicos
- Segregação:
 - Servidores
 - Clusters Kubernetes
 - Databases
 - Filas / Tópicos

Database Bulkheads



Processing Bulkheads

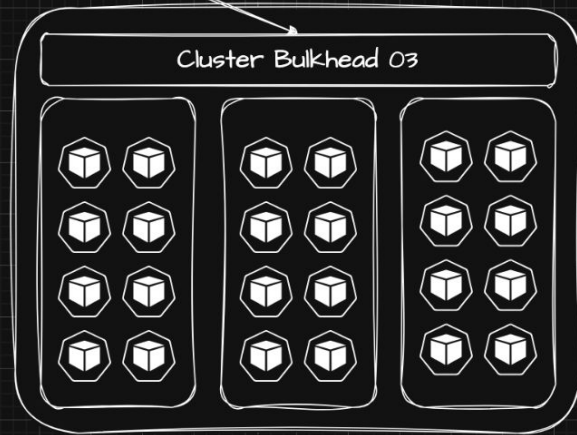
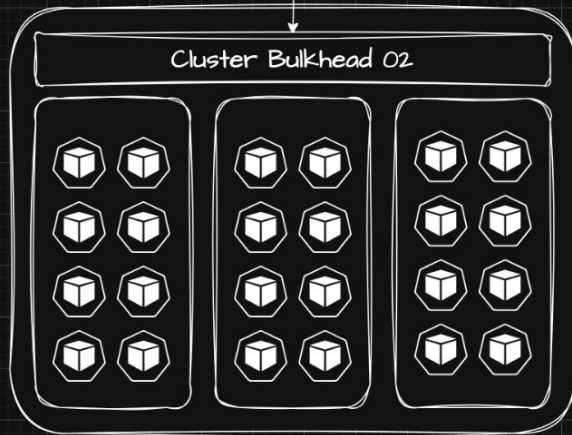
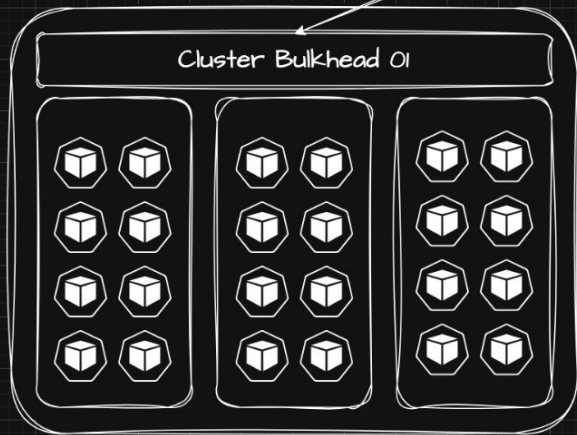


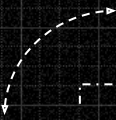
Router / Scheduler

Cluster Bulkhead 01

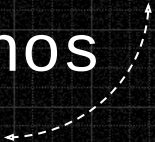
Cluster Bulkhead 02

Cluster Bulkhead 03





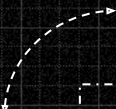
3 Bulkhead e Blast Radius



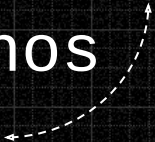
Objetivos e
Ganhos

Bulkhead e Blast Radius

Bulkheads	Blast Radius	Disponibilidade	Impacto
1	100%	0%	Total
2	50%	50%	Muito alto
3	33%	66%	Alto
5	20%	80%	Moderado
10	10%	90%	Moderado
20	5%	95%	Baixo
50	2%	98%	Muito baixo
100	1%	99%	Mínimo



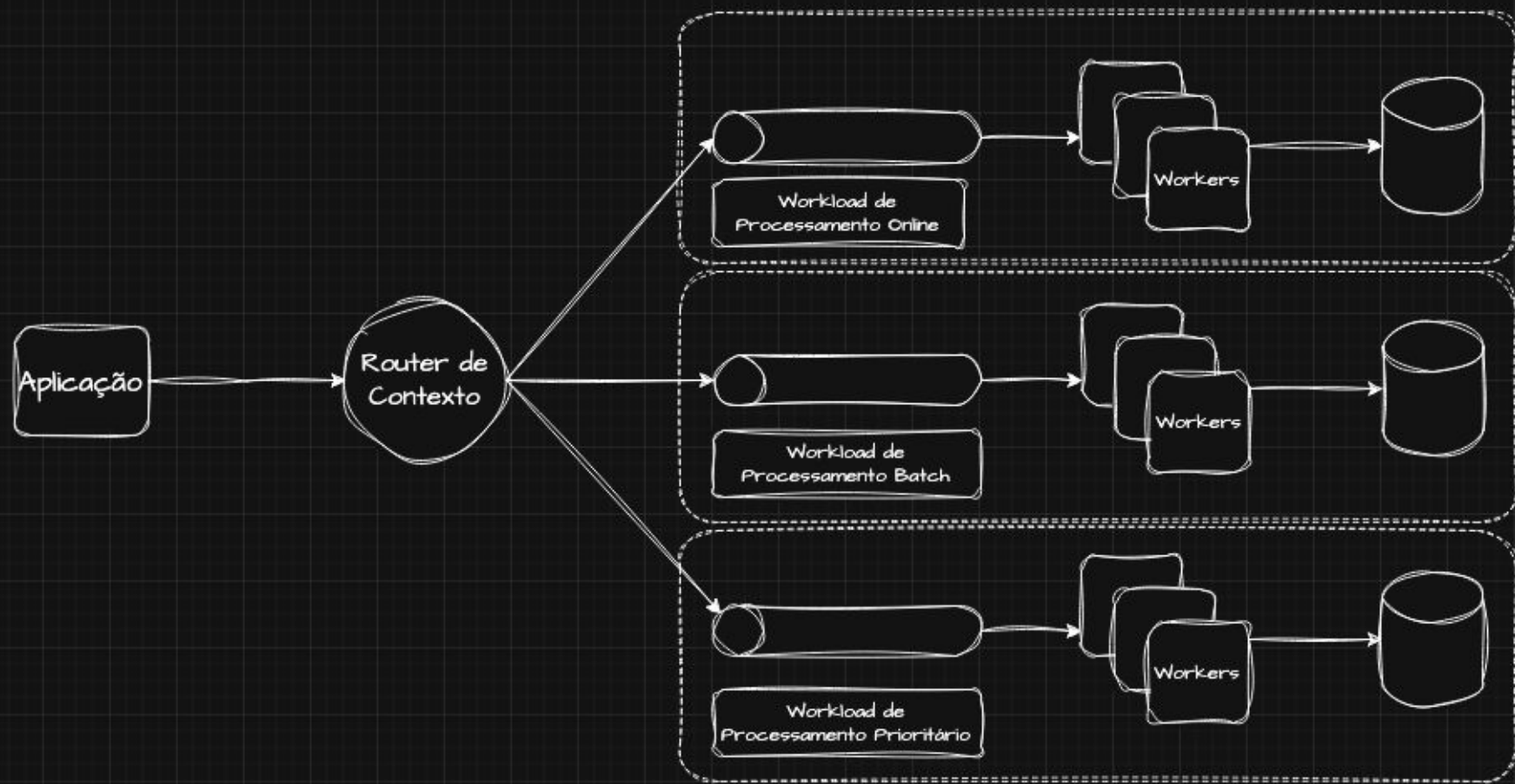
4 Bulkhead Shardings



Objetivos e
Ganhos

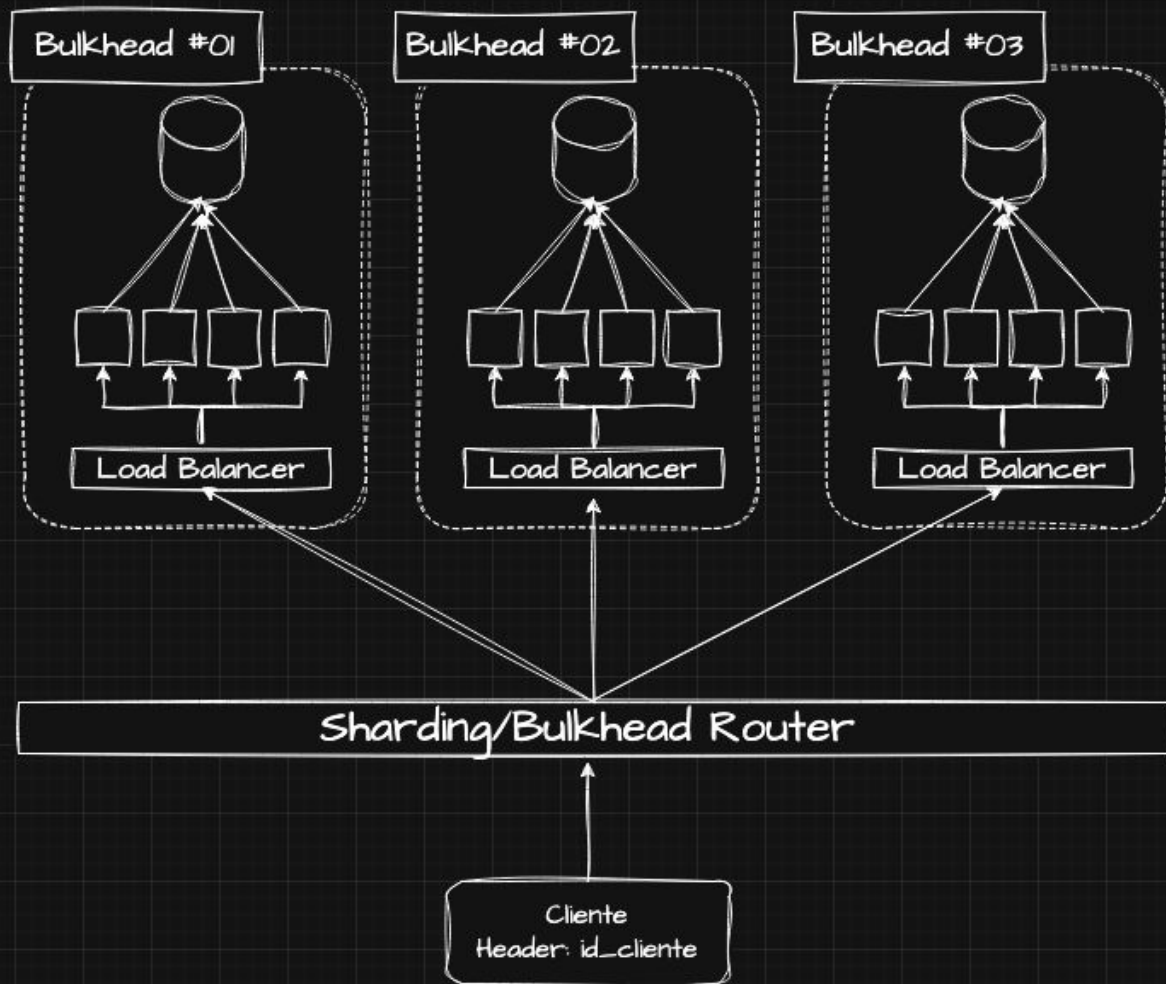
Bulkheads e Sharding Funcional

- Dividido por funcionalidades
- Dividido por padrões de uso
- conjunto específico de funcionalidades
- Recursos próprios e limites bem definidos
- Processamento de pagamentos, consultas e relatórios em shards distintos

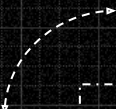


Bulkheads e Sharding Operacional

- Volume ou características de carga
- Sharding por identificadores de cliente, região ou faixa de tráfego
- Limitar o blast radius de picos localizados



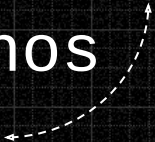
Segregação
Física e
Lógica de
Clientes e
Dados



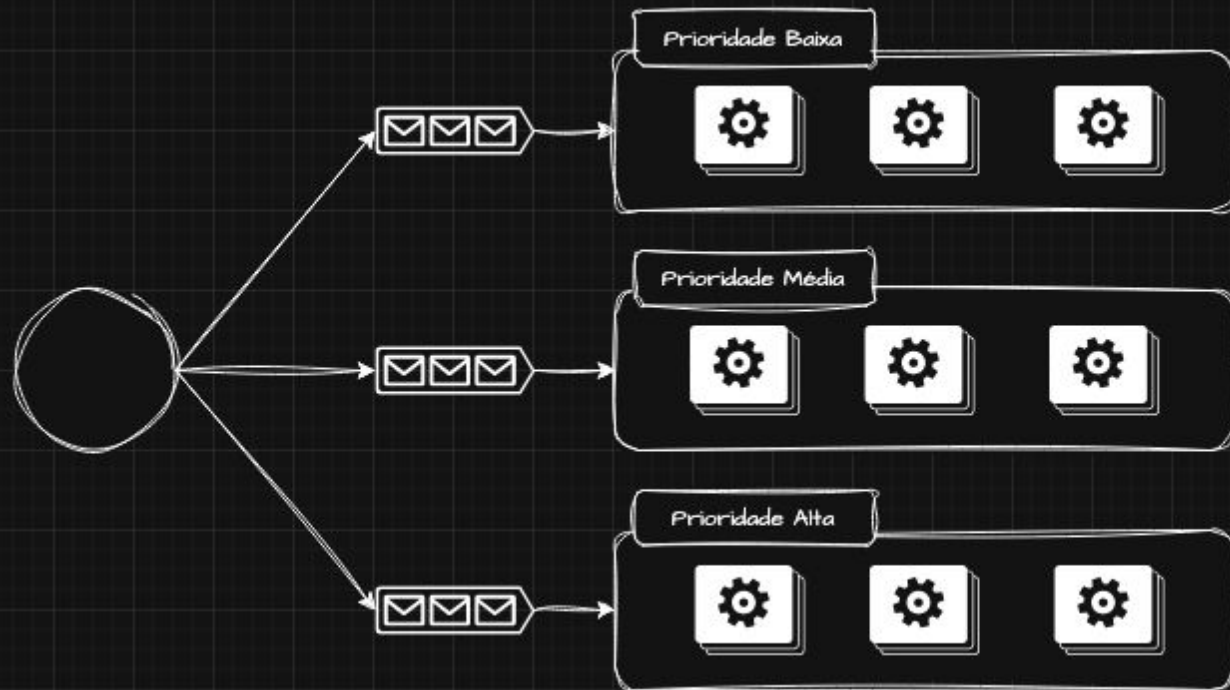
4 Arquiteturas de Bulkheads



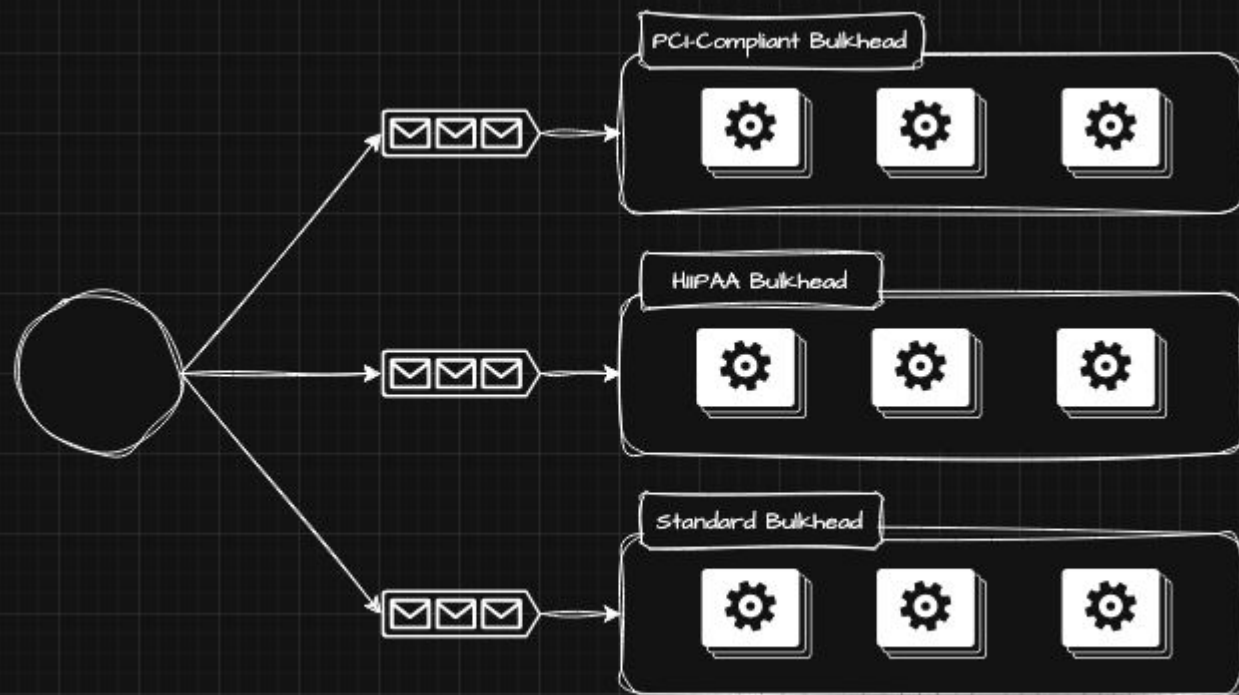
Objetivos e
Ganhos



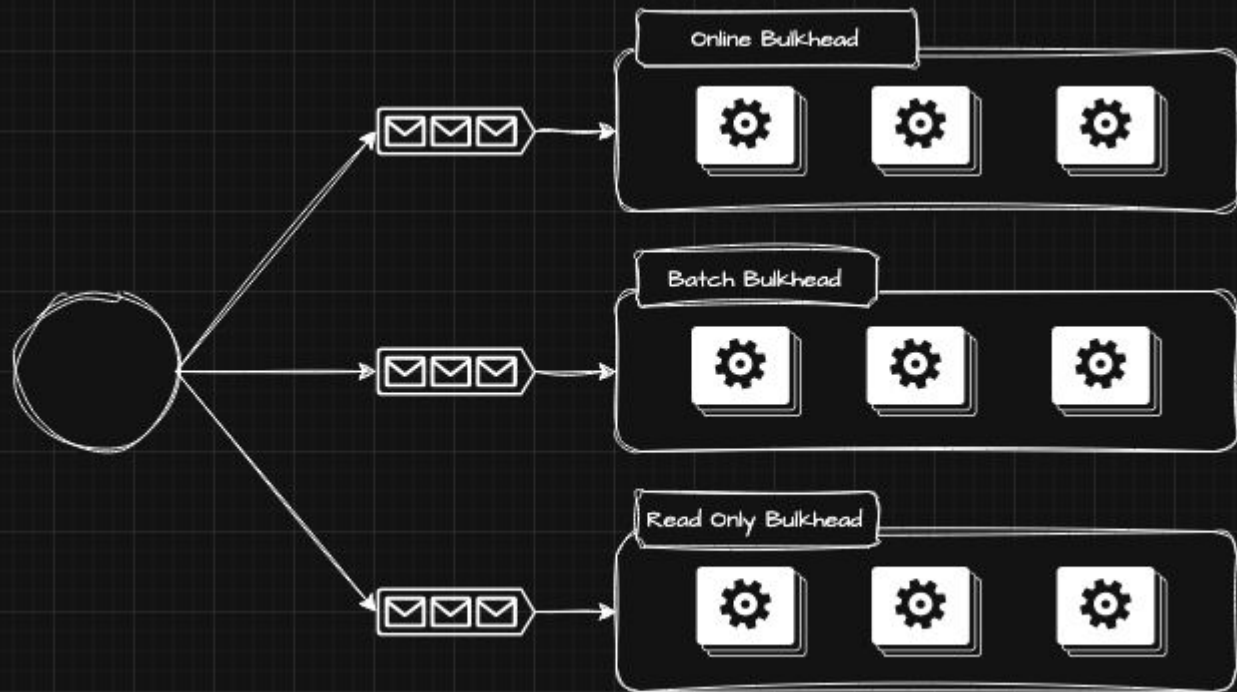
Bulkheads por Priorização



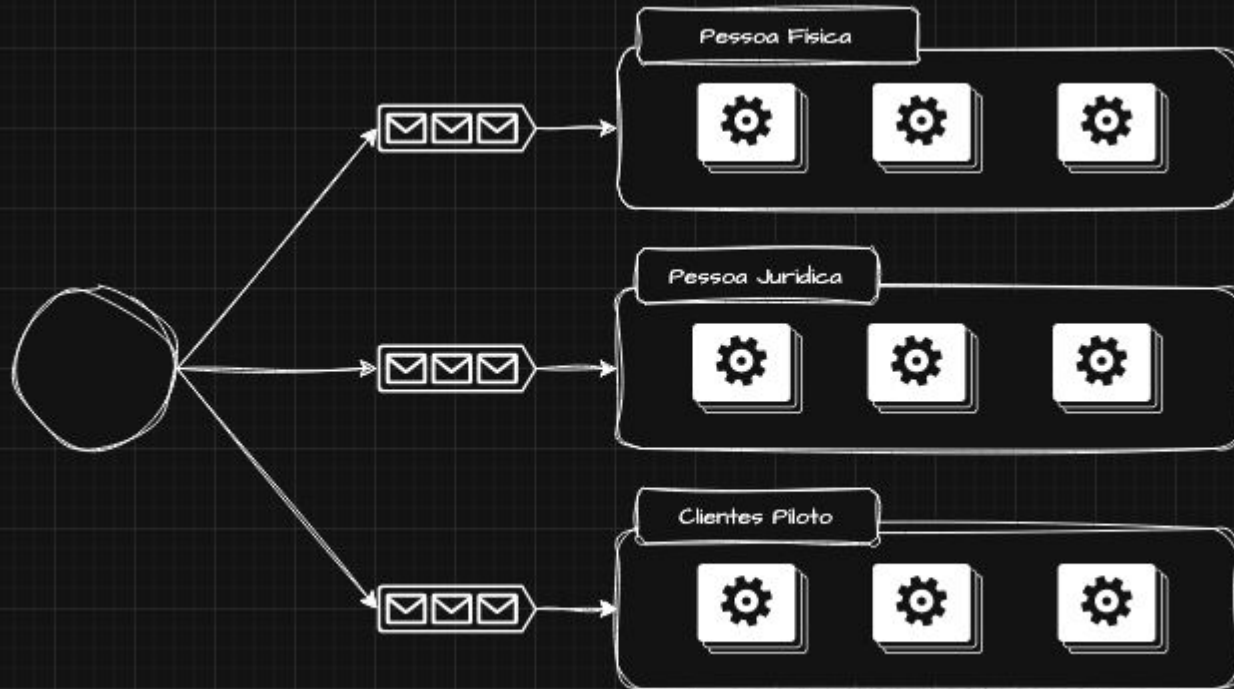
Bulkheads por Criticidade



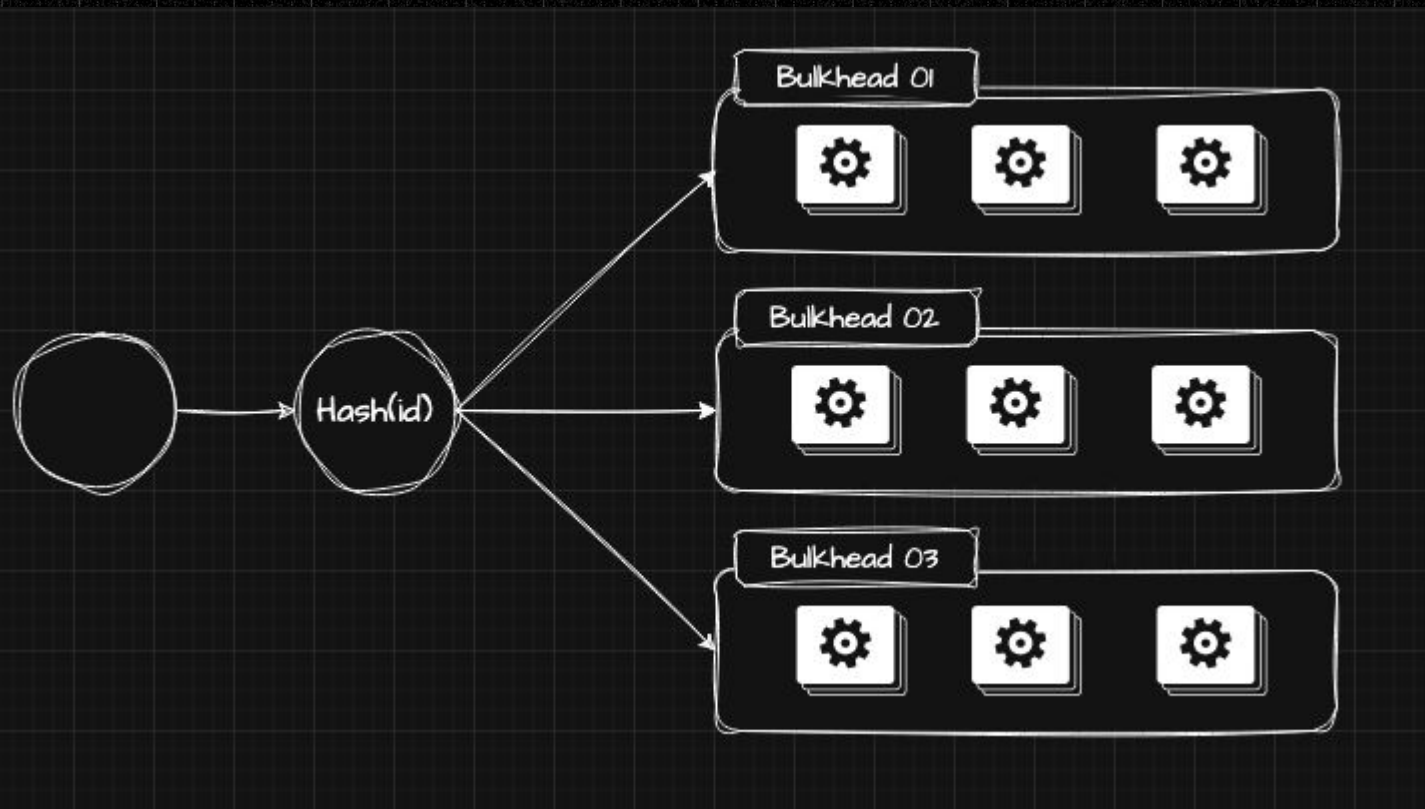
Bulkheads por Tipos de Uso



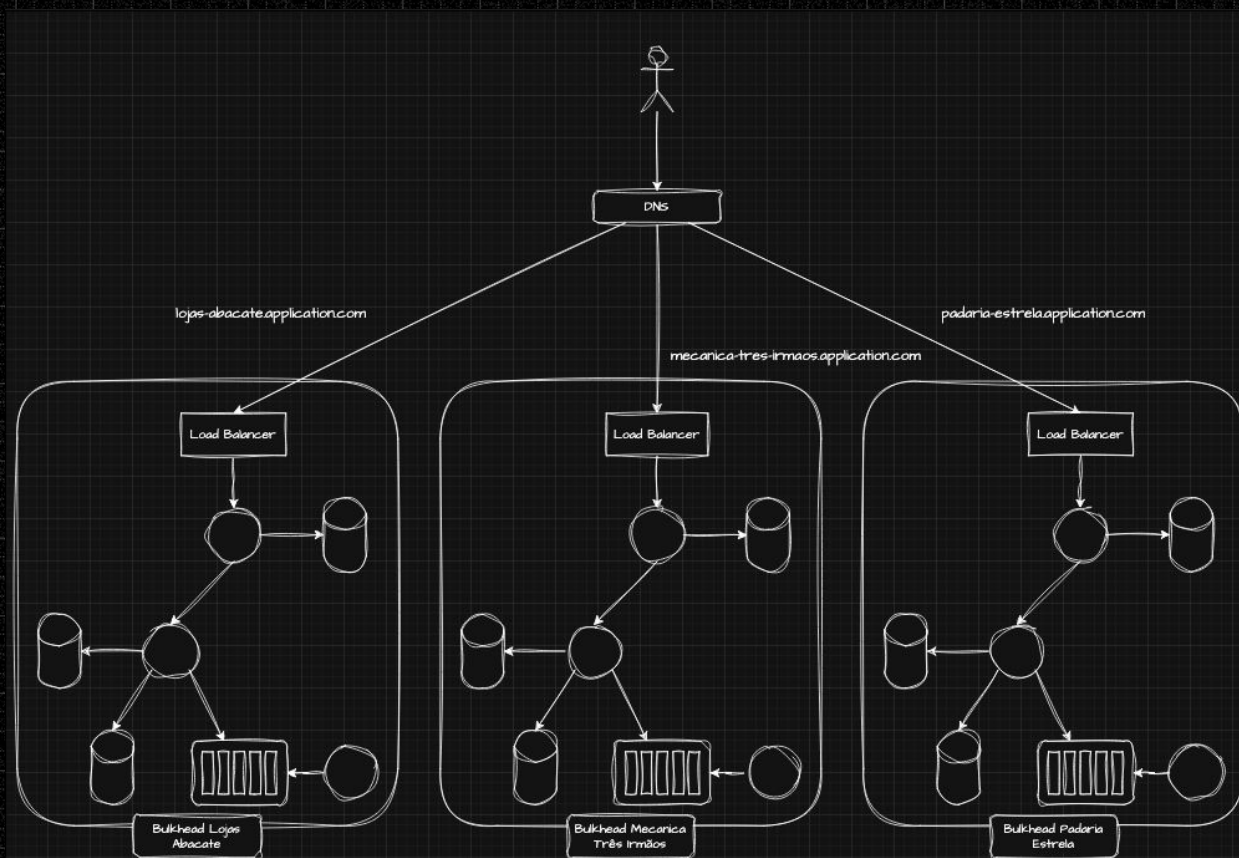
Bulkheads por Segmento



Bulkheads por Hashing Consistente



Bulkheads por Tenants



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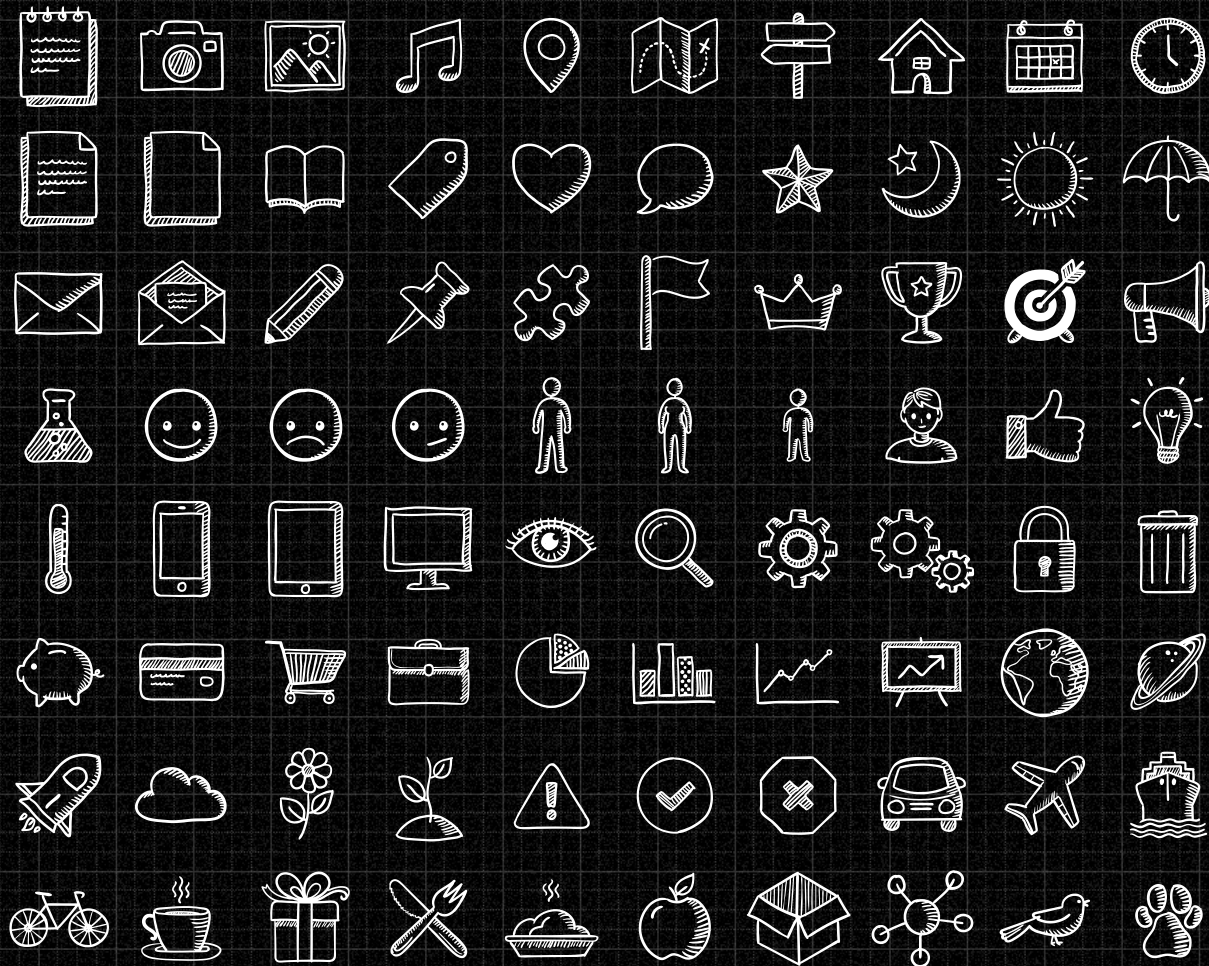
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SlidesCarnival icons are editable shapes.

This means that you can:

- Resize them without losing quality.
- Change fill color and opacity.

Isn't that nice? :)

Examples:

